

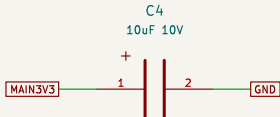
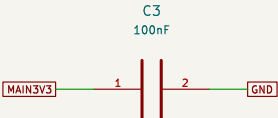
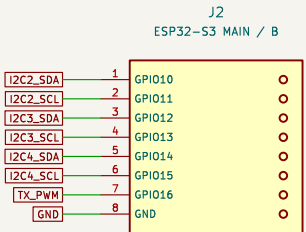
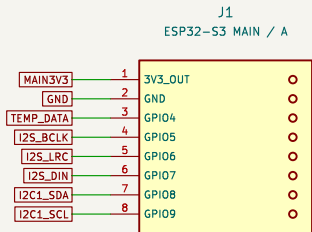
	1	2	3	4	5	6	7	8						
A	AUV V3 integrated TX, receiver and module carrier Solder wires by SIGNAL NAME. Controller and module footprints are intentionally not assumed. All repeated global labels are electrically connected throughout this project. Read the power configuration and assembly notes before connecting USB and external supplies.													
B	Controllers and power interfaces ESP32–S3 main/TX and classic ESP32 RX use separate 3.3 V regulator outputs; both share GND. USB powers controllers; no VIN/VBUS tie. File: controllers.kicad_sch				External 5 V distribution Regulated external 5 V for transmitter, audio and optional sensor supplies. Fuse limit 1 A; test external rail before use. File: power.kicad_sch									
C	Four OLED buses and temperature sensor Four SSD1306 displays at 0x3C on separate I2C buses; optional pull–ups fitted only if modules do not supply suitable pull–ups. File: displays.kicad_sch				ADS1115 modules and sensor inputs ADS1=0x48; ADS2=0x49. TDS/TURB 0–5 V inputs are divided by two and isolated when main ADC supply is invalid. Existing firmware sensor voltages need x2 scaling. File: sensors.kicad_sch									
D	40 kHz TX and I2S audio interfaces A03400A level translator and TC4426A inverting driver give noninverted 5 V gate drive. IRLZ44N low–side switch; 1k piezo reset. MAX98357A external module; GAIN left floating. File: transmitter.kicad_sch				40 kHz receive amplifier MCP6002 follower and nominal gain23 amplifier on RX3V3. No slow 100nF filter at GPIO34; verify 40kHz gain and ADC waveform. File: receiver.kicad_sch									
E														
F	Firmware mapping: main ESP32–S3 GPIO4-16; classic ESP32 receiver ADC input GPIO34. Existing sensor modules and displays connect by labelled pads; retain separate I2C buses as drawn.						Generated from design.json; all symbol geometry is embedded. Functional solder pads; follow pad labels, not module header positions.AUV V3 – module carrierSheet: /File: AUV_V3_Combined.kicad_sch Title: AUV V3 – design overview <table><tr><td>Size: A3</td><td>Date: 2026–09–15</td><td>Rev: B</td></tr><tr><td>KiCad E.D.A. 9.0.7</td><td colspan="2">Id: 1/7</td></tr></table>		Size: A3	Date: 2026–09–15	Rev: B	KiCad E.D.A. 9.0.7	Id: 1/7	
Size: A3	Date: 2026–09–15	Rev: B												
KiCad E.D.A. 9.0.7	Id: 1/7													
	1	2	3	4	5	6	7	8						

Controllers and power interfaces

ESP32-S3 main/TX and classic ESP32 RX use separate 3.3 V regulator outputs; both share GND. USB powers controllers; no VIN/VBUS tie.

Global labels connect sheets • x = Intentionally unused pin • values and pin numbers are authoritative

Circuit connections



Defined power sources / ERC source declarations



Generated from design.json; all symbol geometry is embedded.
Functional solder pads: follow pad labels, not module header positions.
AUV V3 – module carrier
Sheet: /Controllers and power interfaces/
File: controllers.kicad_sch

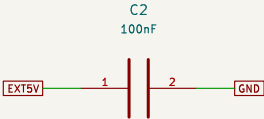
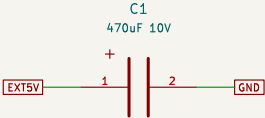
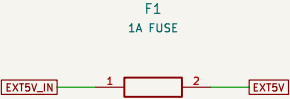
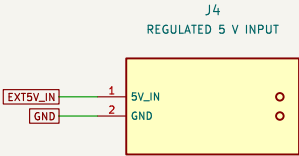
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Size: A3	Date: 2026-09-15	Rev: B
KiCad E.D.A. 9.0.7		
Id: 2/7		

External 5 V distribution

Regulated external 5 V for transmitter, audio and optional sensor supplies. Fuse limit 1 A; test external rail before use.

Global labels connect sheets • x = Intentionally unused pin • values and pin numbers are authoritative

Circuit connections

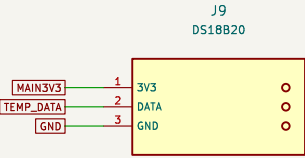
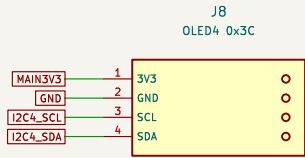
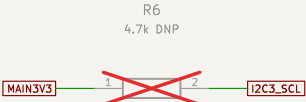
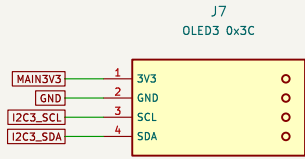
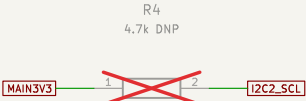
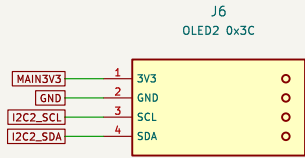
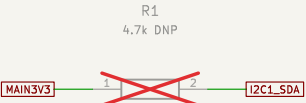
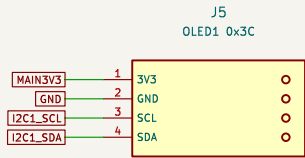


Four OLED buses and temperature sensor

Four SSD1306 displays at 0x3C on separate I2C buses; optional pull-ups fitted only if modules do not supply suitable pull-ups.

Global labels connect sheets • x = Intentionally unused pin • values and pin numbers are authoritative

Circuit connections



Generated from design.json; all symbol geometry is embedded.
Functional solder pads: follow pad labels, not module header positions.
AUV V3 – module carrier
Sheet: /Four OLED buses and temperature sensor/
File: displays.kicad_sch

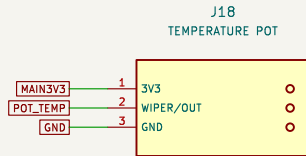
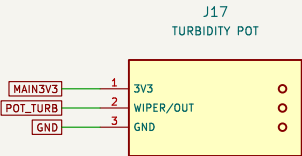
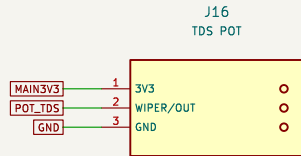
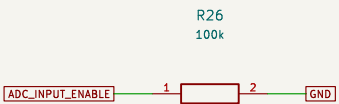
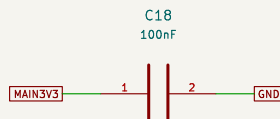
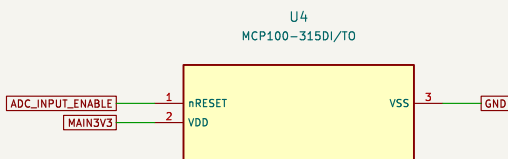
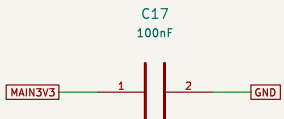
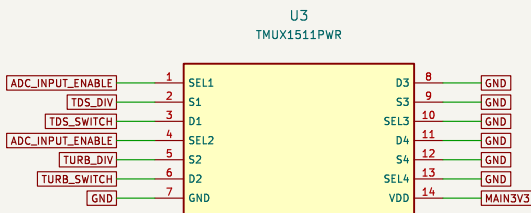
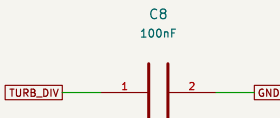
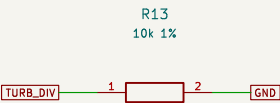
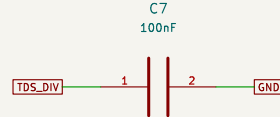
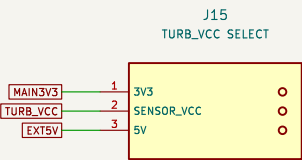
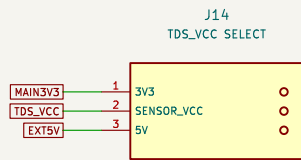
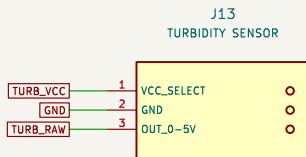
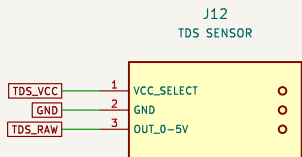
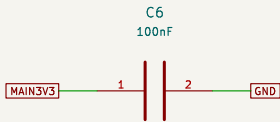
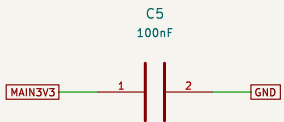
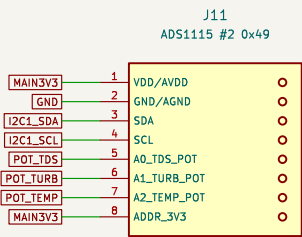
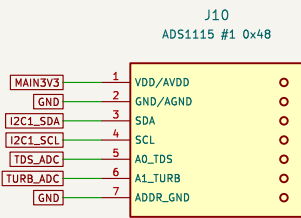
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Size: A3	Date: 2026-09-15	Rev: B
KiCad E.D.A. 9.0.7		Id: 4/7

ADS1115 modules and sensor inputs

ADS1=0x48; ADS2=0x49. TDS/TURB 0–5 V inputs are divided by two and isolated when main ADC supply is invalid.
Existing firmware sensor voltages need x2 scaling.

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Circuit connections



Generated from design.json; all symbol geometry is embedded.
Functional solder pads; follow pad labels, not module header positions.

AUV V3 – module carrier

Sheet: /ADS1115 modules and sensor inputs/
File: sensors.kicad_sch

Title: ADS1115 modules and sensor inputs

Size: A2 Date: 2026-09-15

Rev: B

KiCad E.D.A. 9.0.7

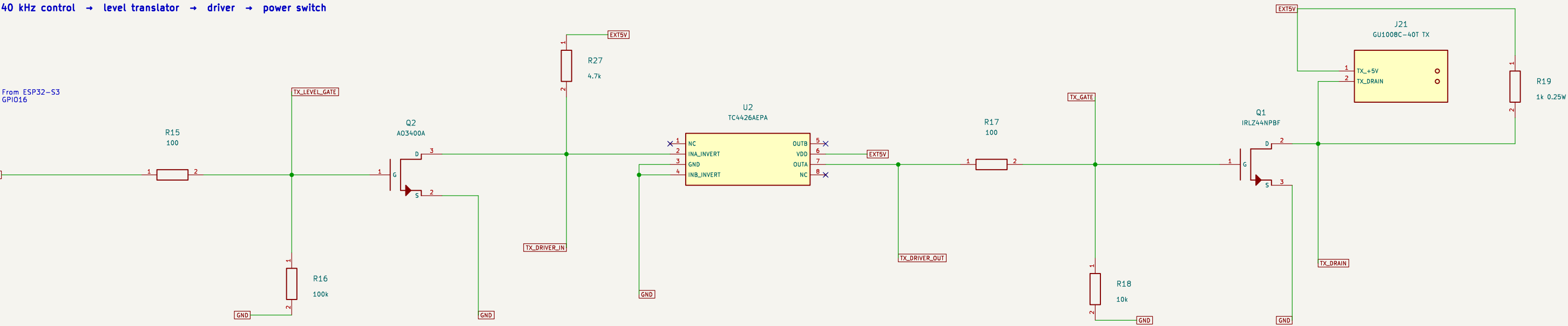
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40 kHz TX and I2S audio interfaces

A03400A level translator and TC4426A inverting driver give noninverted 5 V gate drive. IRLZ44N low-side switch; 1k piezo reset. MAX98357A external module; GAIN left floating.

Global labels connect sheets • x = Intentionally unused pin • values and pin numbers are authoritative

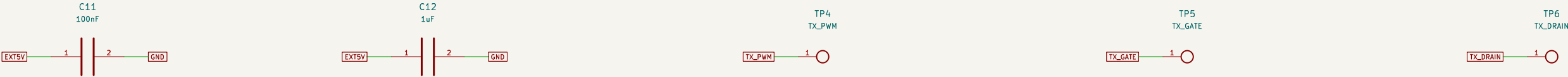
40 kHz control → level translator → driver → power switch



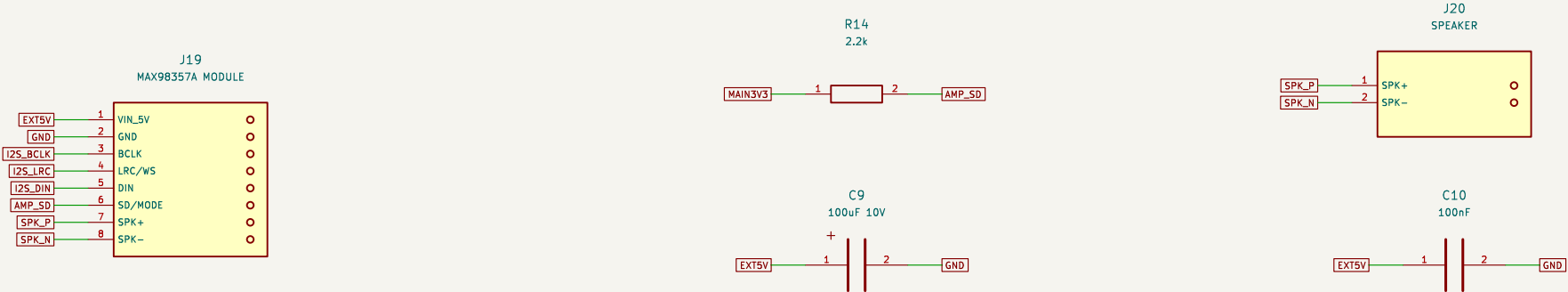
Q2 and U2 each invert; the final gate follows GPIO16. R27 restores the driver input high when Q2 is off.

R19 provides the piezo reset path.

Local driver bypass and TX measurement points



Separate I2S audio module and speaker



Speaker connects only between SPK+ and SPK-. Neither terminal connects to GND. MAX98357A GAIN stays floating.

Generated from design.json; all symbol geometry is embedded. Functional solder pads: follow pad labels, not module header positions.

AUV V3 – module carrier

Sheet: /40 kHz TX and I2S audio interfaces/
File: transmitter.kicad_sch

Title: 40 kHz TX and I2S audio interfaces

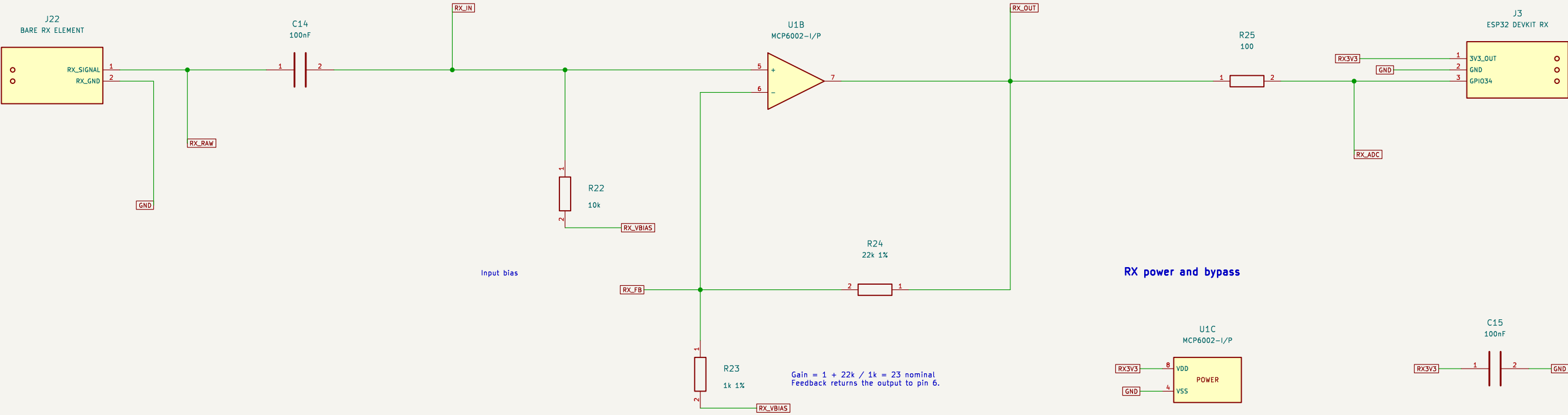
Size: A3	Date: 2026-09-15	Rev: B
KiCad E.D.A. 9.0.7		Id: 6/7

40 kHz receive amplifier

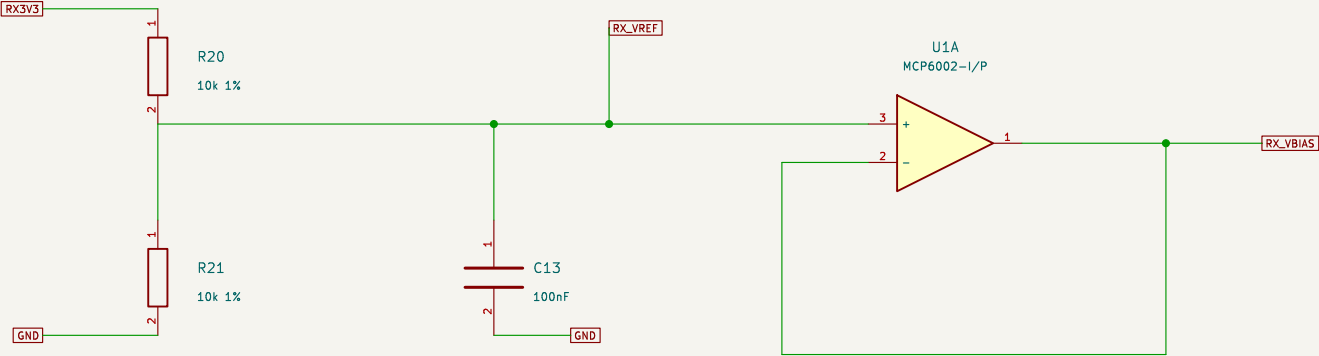
MCP6002 follower and nominal gain23 amplifier on RX3V3. No slow 100nF filter at GPIO34; verify 40kHz gain and ADC waveform.

Global labels connect sheets • x = Intentionally unused pin • values and pin numbers are authoritative

Receiver signal → amplify around 1.65 V → ESP32 ADC

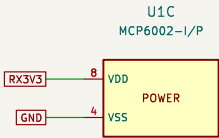


Reference divider and buffer

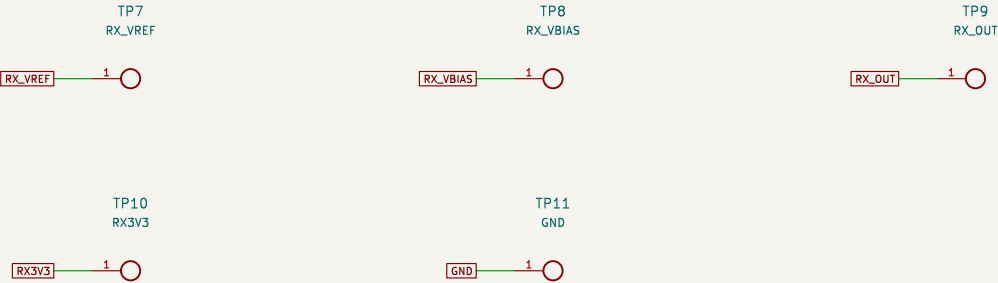


Two equal resistors make half the 3.3 V supply. U1A holds this reference steady for the amplifier.

RX power and bypass



Measurement points



Generated from design.json; all symbol geometry is embedded.
Functional solder pads: follow pad labels, not module header positions.

AUV V3 – module carrier

Sheet: /40 kHz receive amplifier/
File: receiver.kicad_sch

Title: 40 kHz receive amplifier

Size: A3	Date: 2026-09-15	Rev: B
KiCad E.D.A. 9.0.7		Id: 7/7